

VCP: Single-Stage Cosine Classifier and Prototype Learning for Few-Shot Object Detection

1st Xiang Fei
College of Engineering
Qufu Normal University
Rizhao, China
Fly39@qfnu.edu.cn

2nd Mengru Li
College of Engineering
Qufu Normal University
Rizhao, China
13864085521@qfnu.edu.cn

3rd Jinming Huang^{*}
College of Engineering
Qufu Normal University
Rizhao, China
huangjm@qfnu.edu.cn

4th Jacob Liu
Technology Department
GoerTek Inc.
Weifang, Shandong, China
2374622833@qq.com

5th Xiaofu Zhang
Technology Department
Shandong Free-Optics Technology Co., Ltd.
Weifang, China
fly39@icloud.com

6th Xiaojiao Xu
Technology Department
Shandong Huimi Intelligent Technology Co., Ltd.
Rizhao, China
17604705924@163.com

Abstract—Few-shot object detection (FSOD) remains predominantly built upon two-stage architectures such as Faster R-CNN, while YOLO-series single-stage detectors—dominant in industrial deployment—lack systematic FSOD investigation. We present VCP (Visual-Cosine-Prototype), combining YOLO11s with a cosine classifier and prototype learning under the TFA protocol. On PASCAL VOC, VCP achieves 72.2% mAP50 (3-split mean) at 10-shot with only 9.5M parameters ($\sim 1/6$ of competing methods), yet underperforms standard fine-tuning on MS COCO. Through P-R decomposition across 24 comparison groups, we reveal that the cosine classifier acts as a *Recall Booster*, consistently shifting predictions toward higher recall at the expense of precision. We formulate the P-R Tradeoff Hypothesis: net benefit depends on baseline recall headroom and the dataset-specific P-R tradeoff ratio, explaining the VOC-COCO divergence. We also explore Florence-2 semantic descriptions as a lightweight auxiliary prior (+0.5 pp, zero VLM inference overhead). VCP runs at 54.4 FPS, 4–5 $\times$ faster than two-stage FSOD methods.

Index Terms—few-shot object detection, single-stage detector, cosine classifier, prototype learning, recall booster

I. INTRODUCTION

Few-shot object detection (FSOD) aims to detect novel categories from only a few annotated examples per class. Methods such as FSCE [1] and DeFRCN [2] have advanced FSOD benchmarks, yet virtually all are built upon the two-stage Faster R-CNN [3], whose RPN provides built-in foreground filtering that effectively controls false positives under data scarcity. In industrial deployment, however, the YOLO family [4], [5] dominates due to real-time efficiency. This creates a disconnect where “published methods cannot be deployed, and deployed systems lack research foundations.” No prior work has systematically studied modern YOLO detectors (v8+) for FSOD under the TFA protocol.

Supported in part by Key R&D Program of Shandong Province, China (2025TSGCXTHG031) and National Key R&D Program of China (2021YFE0193900). Corresponding author: Jinming Huang (huangjm@qfnu.edu.cn).

The core challenge for single-stage FSOD is the absence of RPN filtering: dense prediction generates abundant low-quality proposals, subjecting the classifier to extreme positive-negative imbalance. The cosine classifier [6] mitigates overfitting through normalization, while Florence-2 [8] offers fine-grained semantic understanding. However, combining cosine classification with single-stage YOLO—and systematically analyzing its behavior—remains unexplored.

We propose VCP, replacing YOLO11s’ linear classification head with a cosine classifier initialized by support-set visual prototypes (Proto mode), and further integrating Florence-2 semantic descriptions as auxiliary priors (Fused mode). On VOC, VCP achieves 72.2% mAP50 at 10-shot (3-split mean) with 9.5M parameters, competitive with 60M-parameter SOTA methods; Split 1 reaches 74.7%, surpassing all existing two-stage methods.

Strikingly, the same approach underperforms fine-tuning on COCO. Through systematic P-R decomposition across 24 groups, we identify the cosine classifier as a **Recall Booster** that invariably sacrifices precision for recall. We formulate the **P-R Tradeoff Hypothesis** to explain this cross-dataset divergence. Our contributions are: (1) the first systematic exploration of single-stage YOLO under TFA FSOD; (2) identification of the Recall Booster and P-R Tradeoff Hypothesis; (3) VLM semantic prototypes with zero inference overhead; and (4) per-class temperature scaling to partially counteract the Recall Booster’s precision loss.

II. RELATED WORK

FSOD has transitioned from early meta-learning [9], [10] to the fine-tuning paradigm of TFA [6]. FSCE [1], DeFRCN [2], and KD-DeFRCN [12] improved proposal encoding, gradient decoupling, and distillation. Multi-scale augmentation [13], retrieval-based adaptation [14], and margin optimization [15] further advanced the field. All these methods share the Faster R-CNN backbone.

VCP (VLM-Cosine-Prototype) Architecture for Few-Shot Object Detection

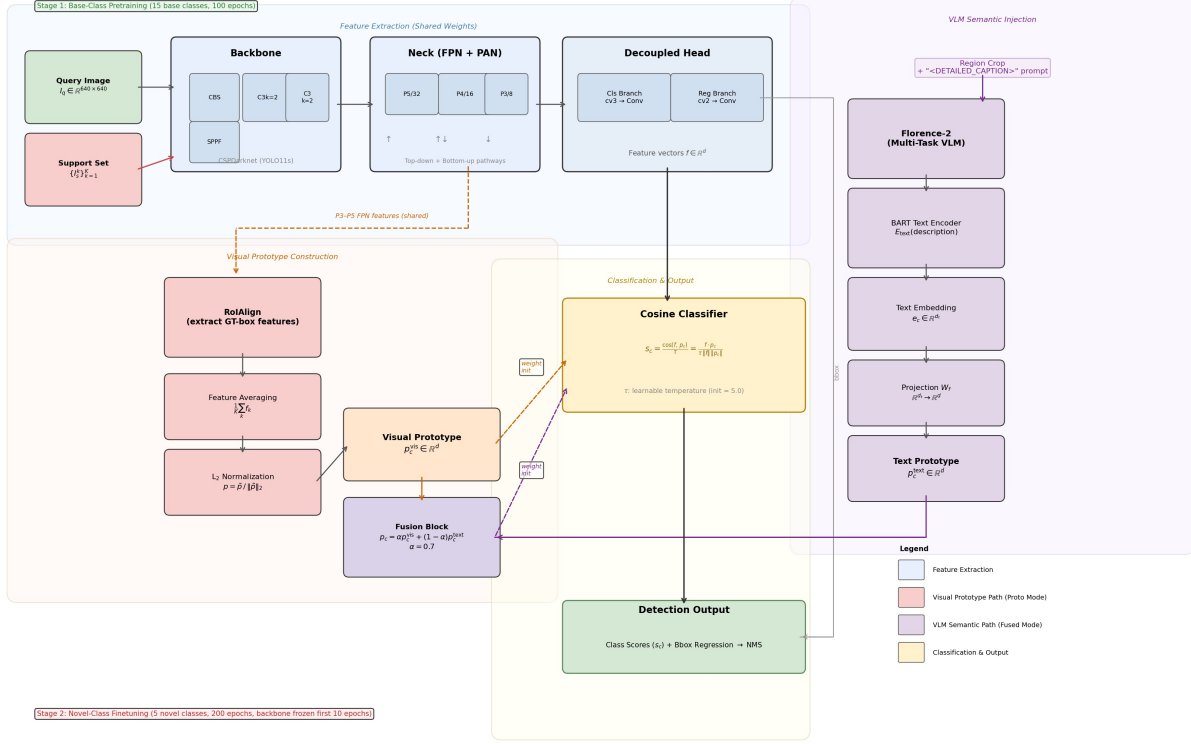


Fig. 1. VCP framework: YOLO11s backbone+neck, decoupled head, cosine classifier. Visual prototypes via RoIAlign; semantic prototypes via Florence-2 + BART. Fused at $\alpha = 0.7$.

Single-stage FSOD exploration remains extremely limited. Meta-YOLO used meta-learning over YOLOv2 with high-variance splits; recent designs [11] explore contrastive learning under TFA. No prior work studies modern YOLO (v8+) under TFA. Our semantic branch is not an architectural novelty but rather a strategic choice: few-shot visual prototypes are inherently noisy, and detailed VLM captions provide complementary priors with zero inference overhead.

VLMs such as CLIP [7] and Florence-2 [8] enable semantic prototype construction, while OWL-ViT [16], GLIP [17], and Grounding DINO [18] have advanced open-vocabulary detection. Leveraging VLM prototypes under FSOD’s strict constraints (1–10 samples) remains underexplored.

III. METHOD

A. Framework and Prototypes

VCP builds upon YOLO11s, replacing its linear classification head with a cosine-similarity classifier (Fig. 1). For a proposal feature $f \in \mathbb{R}^d$, the score for class c is $s_c = \cos(f, p_c) / \tau$, where p_c is the class prototype and τ a learnable temperature. L2-normalization ($\|f\| = \|p_c\| = 1$) ensures classification depends solely on directional alignment, mitigating overfitting.

Prototypes are derived from two sources. **Visual Proto-type (Proto)**: For each novel class c , RoIAlign extracts

features within GT boxes from FPN P3–P5, averaged and L2-normalized: $p_c^{\text{vis}} = \frac{1}{K} \sum_{i=1}^K \text{RoIAlign}(F_i, b_i)$, written directly into classifier weights.

Semantic Prototype (Fused): Visual prototypes from only 5–10 support images inherently suffer from high variance—viewpoint bias, occlusion, and lighting variation cannot capture full class diversity. We introduce semantic prototypes as complementary priors: Florence-2 generates `<DETAILED_CAPTION>` descriptions capturing fine-grained visual attributes beyond what few-shot images convey, and BART encodes these into p_c^{text} . The fused prototype is:

$$p_c = \alpha \cdot p_c^{\text{vis}} + (1 - \alpha) \cdot W_f \cdot p_c^{\text{text}} \quad (1)$$

where $\alpha = 0.7$ and W_f is a learnable projection. Critically, semantic extraction occurs only during training; at inference, prototypes are fixed, incurring zero VLM overhead.

B. Recall Booster and Per-Class Temperature

The gradient of cosine similarity reveals its behavioral mechanism:

$$\frac{\partial \cos(f, p_c)}{\partial f} = \frac{p_c - \cos(f, p_c) \cdot \hat{f}}{\|f\|}, \quad \hat{f} = \frac{f}{\|f\|} \quad (2)$$

When f has low similarity to p_c , gradients are large, driving the boundary outward; when similarity is high, gradients diminish. Unlike linear classifiers whose gradients are uniform,

TABLE I
VOC MAP50 (SPLIT 1 / 3-SPLIT MEAN $\pm$ STD)

Method	1-shot	3-shot	5-shot	10-shot
Finetune	10.9 / –	20.5 / –	32.9 / –	51.3 / –
Cosine	12.8 / –	51.3 / –	66.9 / 57.8 $\pm$ 2.9	70.7 / 69.0 $\pm$ 3.1
Cosine+Proto	21.5 / –	52.3 / –	68.8 / 57.8 $\pm$ 2.8	74.4 / 71.7 $\pm$ 2.7
Cosine+Fused	21.3 / –	53.9 / –	69.0 / 58.1 $\pm$ 2.5	74.7 / 72.2 $\pm$ 2.6

the cosine classifier *disproportionately updates low-confidence samples*, favoring high recall at the cost of precision. We term this the **Recall Booster**.

To partially counteract this precision loss, we replace the single global temperature τ with per-class temperatures τ_c :

$$s_c = \tau_c \cdot \cos(f, p_c) + b_c \quad (3)$$

Classes prone to false positives learn higher τ_c , flattening their similarity distribution and suppressing overconfident incorrect predictions. This adds only C extra parameters (e.g., 20 for VOC) with negligible overhead.

IV. EXPERIMENTS

A. Setup

We evaluate on PASCAL VOC 2007+2012 and MS COCO 2014 under the TFA split protocol [6]: VOC 20 classes partitioned into 15 base / 5 novel (three splits); COCO 20 VOC-overlapping classes as novel, 60 as base. The base detector is YOLO11s (9.5M parameters), pre-trained on COCO base classes. Stage 1: 100 epochs on base classes (batch 64, lr 0.01). Stage 2: fine-tune on novel classes up to 200 epochs with early stopping (patience 30, batch 4, lr 0.0005). SGD optimizer (momentum 0.937, weight decay 0.0005, 3-epoch warmup), backbone frozen first 10 epochs. Augmentation: mosaic, RandAugment, HSV jitter, random flip/scale/translate, random erasing. Input 640×640 , τ initialized at 5.0. All experiments use a single NVIDIA H20 GPU.

Four variants compared: (1) *Finetune*—standard YOLO11s with linear classifier; (2) *Cosine*—cosine classifier, random init; (3) *Cosine+Proto*—cosine classifier with visual prototypes; (4) *Cosine+Fused*—cosine classifier with fused visual + Florence-2 prototypes.

B. Main Results

VOC. Table I reports VOC mAP50 (Split 1 / mean $\pm$ std). Three observations: (1) The cosine classifier drives substantial gains—5-shot: 32.9% $\rightarrow$ 66.9% (+34.0 pp); 10-shot: 51.3% $\rightarrow$ 70.7% (+19.4 pp). (2) Visual prototype init contributes +3.7 pp at 10-shot. (3) Florence-2 fusion adds +0.3 pp, consistent across splits.

COCO. Table II shows the opposite: Finetune outperforms all cosine variants. At 30-shot, Cosine+Fused reaches 52.3%, above pure Cosine (49.4%) but far below Finetune (60.3%). Results are on a single split and warrant multi-split verification.

TABLE II
COCO MAP50

Method	10-shot	30-shot
Finetune	59.0	60.3
Cosine	47.1	49.4
Cosine+Proto	47.2	51.6
Cosine+Fused	47.0	52.3

C. P-R Analysis

To explain the VOC-COCO divergence, we decompose P-R across all 24 method $\times$ shot $\times$ dataset combinations (Table III). Across all 24 cosine variants, recall increases while precision decreases—a systematic property. On VOC, $\Delta R / |\Delta P| \approx 2.0$ at 5-shot; recall gain dominates. On COCO, the ratio collapses to 0.48 (10-shot), with ΔP far exceeding ΔR . Fused mode partially recovers precision ($P=0.336$ vs. 0.200 for pure Cosine at 10-shot) with minimal recall loss, suggesting semantic priors moderate prototype over-relaxation.

We formalize the **P-R Tradeoff Hypothesis**: the cosine classifier inherently pushes toward high-recall, low-precision operation; net benefit is the product of (1) baseline recall headroom and (2) the dataset P-R tradeoff ratio, determined by class count and scene complexity.

TABLE III
P-R DECOMPOSITION SUMMARY

Setting	ΔP	ΔR	ΔmAP	$\Delta R / \Delta P $
VOC 5-shot Cos+Fused	−0.22	+0.45	+35.9	2.00
VOC 10-shot Cos+Fused	−0.35	+0.33	+23.4	0.96
COCO 10-shot Cos+Fused	−0.81	+0.39	−12.0	0.48
COCO 30-shot Cos+Fused	−0.56	+0.14	−8.0	0.25

D. Ablation and SOTA

Ablation. Table IV shows the cosine head drives $\sim 90\%$ of gains. Visual prototypes (74.4%) outperform Florence-2 text-only (72.1%), and fusion achieves the optimum (74.7%). **Per-class temperature** (VOC 10-shot, Cosine+Fused): Precision improves from 0.01 to 0.123 (12.3 $\times$), with recall decreasing marginally (0.901 $\rightarrow$ 0.875). mAP50-95 shifts from 52.22 to 51.6, showing per-class temperature rebalances the P-R frontier.

SOTA Comparison. Table V compares VCP against existing FSOD methods (VOC Split 1). All baselines use Faster R-CNN R-101 ($\sim 60M$); VCP uses YOLO11s (9.5M). At 5-shot and 10-shot, VCP matches or surpasses all prior methods at 1/6

TABLE IV
ABLATION: COMPONENT & PROTOTYPE SOURCE (VOC)

Configuration	3-shot	5-shot	10-shot
Finetune (baseline)	20.5	32.9	51.3
+ Cosine head	51.3	66.9	70.7
+ Visual Proto	52.3	68.8	74.4
+ VLM Fused	53.9	69.0	74.7
Florence-2 Only (10-shot)	—	—	72.1

the parameters. The 1-shot gap indicates single-stage detectors need additional regularization under extreme scarcity.

TABLE V
COMPARISON WITH SOTA (VOC SPLIT 1, MAP50)

Method	1-shot	3-shot	5-shot	10-shot
TFA w/cos	39.8	44.7	55.7	56.0
FSCE	44.2	51.4	61.9	63.4
DeFRCN	53.6	61.5	64.1	60.8
KD-DeFRCN	58.2	65.1	68.2	67.4
VCP Cos+Fused	21.3	53.9	69.0	74.7
VCP Cos+Proto	21.5	52.3	68.8	74.4

E. Inference Speed

Table VI compares speed (H20, 640×640 , batch 1). VCP achieves 54.4 FPS—4–5 $\times$ faster than two-stage methods—with 9.5M parameters. The cosine classifier adds ~ 4 ms, still well above 30 FPS. Notably, Florence-2 is not loaded at inference; semantic prototypes are pre-computed.

TABLE VI
INFERENCE SPEED (H20, 640×640 , BS=1)

Model	Latency (ms)	FPS	Params
FRCN R-101 (ref.)	~ 85	~ 12	$\sim 60M$
YOLO11s baseline	14.4	69.5	9.5M
Finetune	14.7	68.2	9.5M
Cosine	18.9	52.8	9.5M
Fused (VCP)	18.4	54.4	9.5M

V. CONCLUSION

We presented VCP, combining YOLO11s with cosine classification and prototype learning for single-stage FSOD. Key findings: (1) Single-stage FSOD is viable—VCP achieves competitive performance at 1/6 the parameters of two-stage methods, running at 54.4 FPS. (2) Through 24-group P-R decomposition, we identified the Recall Booster and formulated the P-R Tradeoff Hypothesis, providing a principled explanation of VOC-COCO divergence. (3) Florence-2 semantic prototypes provide complementary priors with zero inference overhead. (4) Per-class temperature scaling partially counteracts the Recall Booster’s precision loss ($12.3\times$ Precision gain, a marginal mAP50-95 decline of -0.62). Limitations include weak 1-shot performance and single-split COCO evaluation. Future work includes adaptive VLM-prototype fusion, cross-domain generalization, and confidence-calibrated detection.

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